THE AIRBNB EFFECT

A Cross-City Analysis of Short-Term Rentals and Residential Housing Markets

Rome · Milan · Florence · New York

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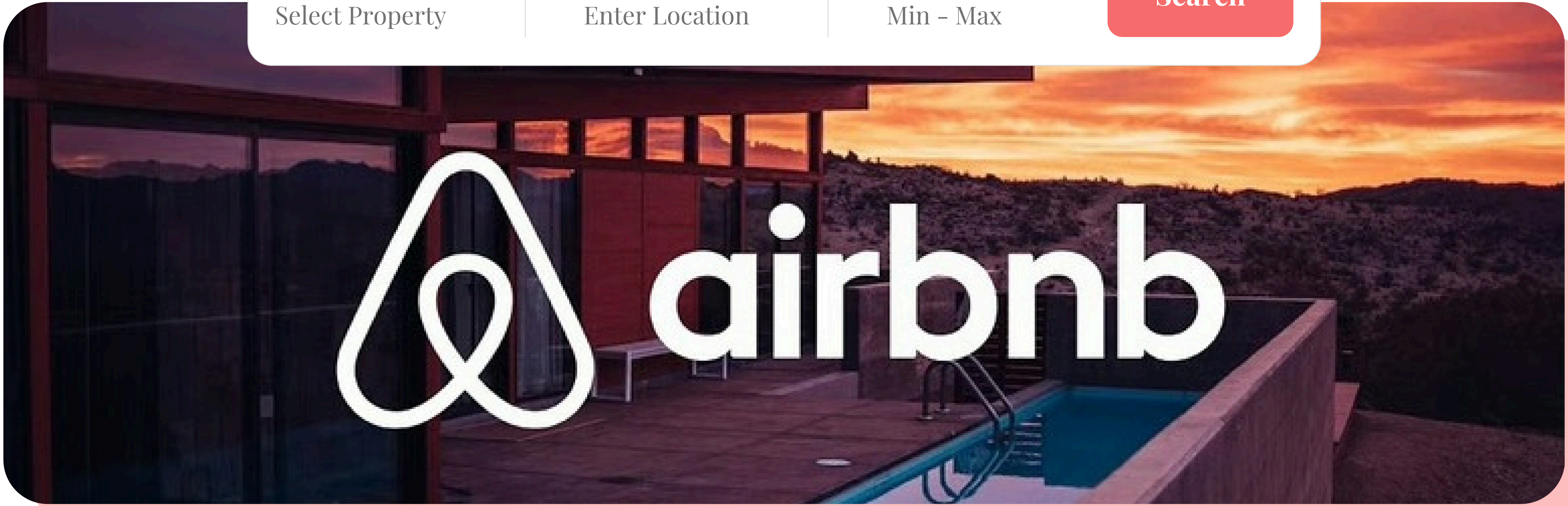


RESEARCH QUESTION

How do short-term rental platforms affect housing markets?

This project studies whether the expansion of Airbnb affects residential rental prices.

The core idea is that **short-term rentals may reduce the supply of housing available to long-term tenants**, creating **upward pressure on rents**.

We test three hypotheses:

- First, that a higher number of Airbnb listings is associated with higher rents.
- Second, that this effect is stronger in highly touristic cities.
- Third, that regulation can effectively reverse this pressure.

Testable Hypotheses

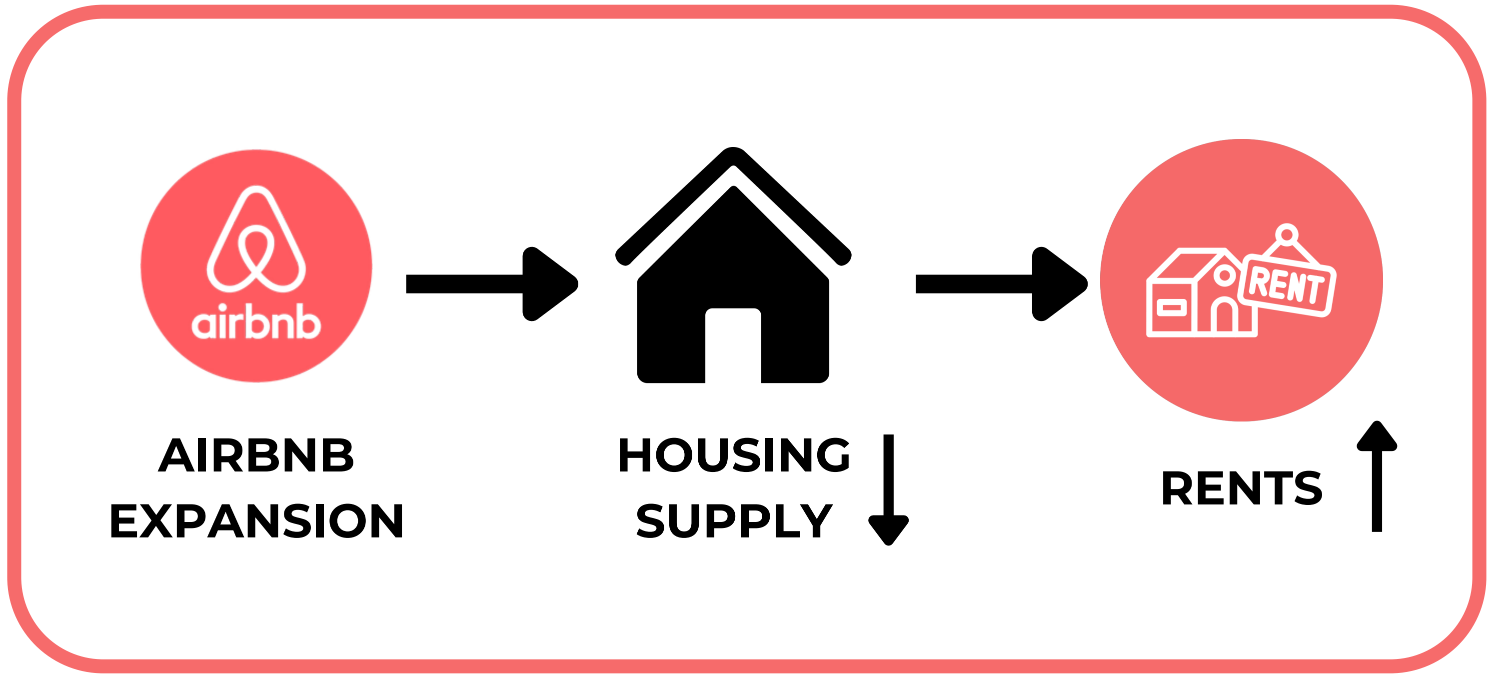
H1: *a higher Airbnb listing density is associated with higher residential rents*

H2: *this effect is stronger in highly touristic / high-demand cities*

H3: *regulatory intervention can effectively reduce this pressure*

EMPIRICAL STRATEGY

Combining two identification strategies



To address this question, we use two complementary empirical approaches.

In **New York**, we exploit a regulatory shock (Local Law 18) which sharply reduced Airbnb listings in September 2023. This allows us to implement an ***Interrupted Time Series*** design and obtain quasi-causal evidence.

In **Italy**, where no comparable shock is available, we estimate a ***cross-sectional OLS*** model using data for Rome, Milan, and Florence, including city and urban-area fixed effects. This approach identifies robust associations, although it does not allow strong causal claims.

DATA SOURCES

This project combines data from multiple sources. Airbnb data is collected from **Inside Airbnb and additional datasets for 2023–2025**.

For rental prices, we use **Zillow's ZORI index** for New York and official **OMI data from the Italian Revenue Agency** for Rome, Milan, and Florence.

These sources differ in structure and units, which is why we estimate separate models for the U.S. and Italian cases.

Inside Airbnb
Adding data to the debate

Agenzia Entrate

Zillow®

The two rent measures (ZORI and OMI) are not directly comparable, so we estimate separate models.

MODEL SPECIFICATION - ITS (NYC)

$$\boxed{Y_t = \alpha + \beta_1 t + \beta_2 D_t + \beta_3 (t - t^*) \cdot D_t + \varepsilon_t}$$

Legend:

- Y_t = Zillow ZORI index ($/month), NYC
- *t* = linear time trend
- D_t = dummy = 1 post Local Law 18 (Sep 2023)
- t* = intervention breakpoint
- β_2 = immediate level shift (main coefficient of interest)
- β_3 = change in post-intervention trend
- Standard errors: OLS standard

A SUDDEN CONTRACTION IN AIRBNB SUPPLY (NYC)

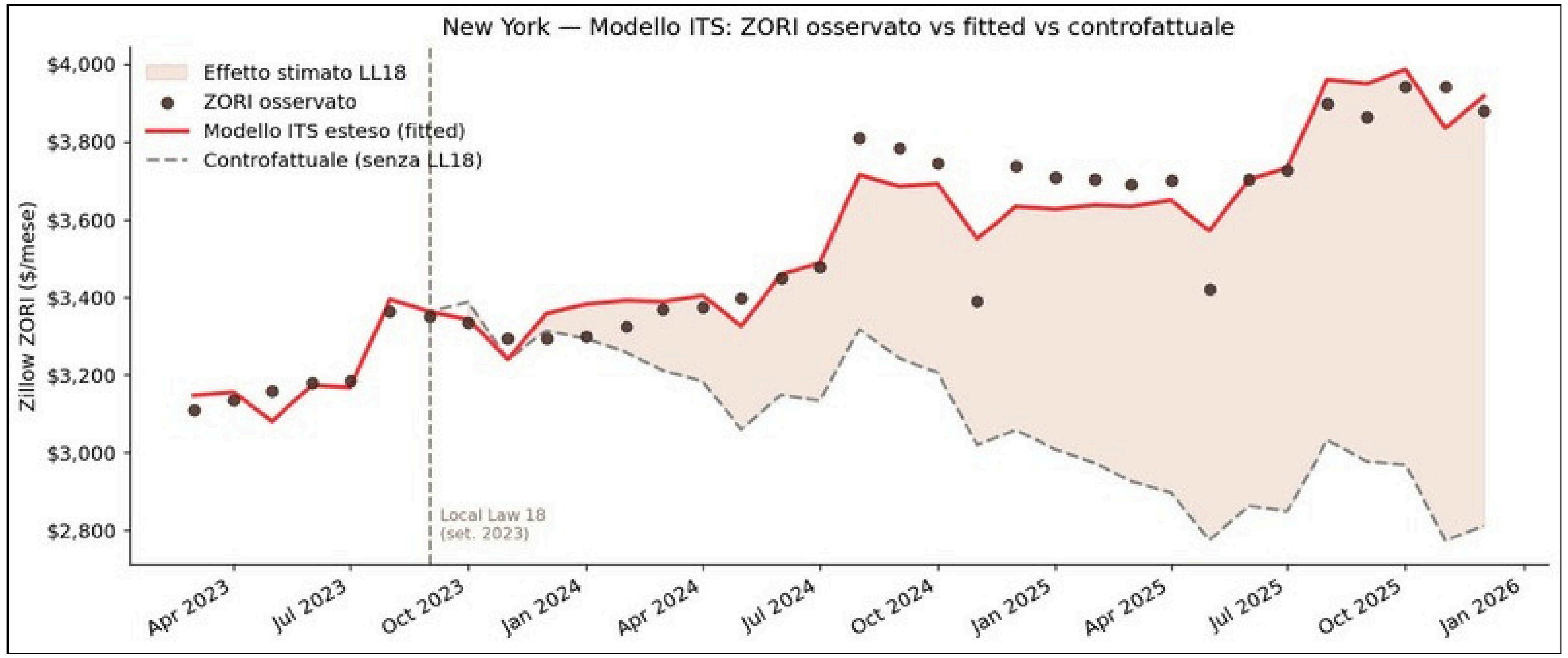
In **September 2023**, New York introduced **Local Law 18**, requiring hosts to register and significantly restricting short-term rentals.

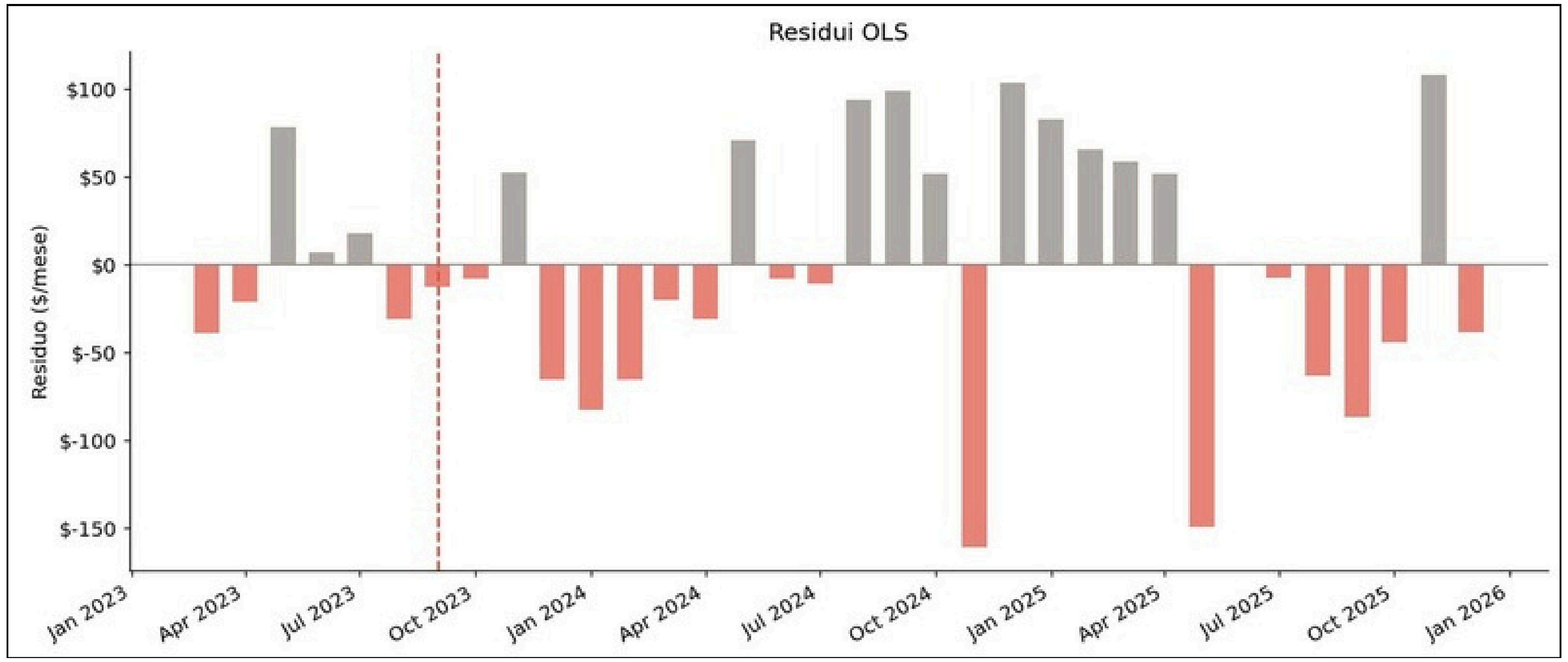
As a result, the **number of active "entire home" listings dropped from approximately 24,000 to 11,500 in just two months**, a **reduction of about 55%**.

This sharp and exogenous contraction in supply provides a unique opportunity to study how rental prices react to a sudden change in the short-term rental market.

MAIN RESULT (NYC)

Rents respond immediately to the supply shock





The Interrupted Time Series analysis shows a **clear and immediate effect of the policy intervention**.

At the time of the regulation, **average rents decreased by approximately 79 to 88 dollars per month**, depending on the specification. This result is remarkably stable across different model variants and represents the most robust finding of the project.

While the post-intervention trend is sensitive to model specification, the immediate drop in rents provides strong **evidence that reducing Airbnb supply can alleviate pressure on the housing market**.

MODEL SPECIFICATION -
OLS Cross-Sectional (ITALY)

$$\text{Rent}_{iz} = \alpha + \beta \log(\text{listings}_{iz}) + \gamma X_{iz} + \delta_c + \delta_f + \varepsilon_{iz}$$

Legend:

- Rent_{iz} = OMI rent (€/m²/month), zona ii i, fascia zz z
- $\log(\text{listings}_{iz})$ = log Airbnb listing density
- X_{iz} = controls: [lista controlli che avete usato]
- δ_c = city fixed effects (Rome, Milan, Florence)
- δ_f = urban area fixed effects (centro, periferia, suburbia, extraurb.)
- β = semi-elasticity (main coefficient of interest)
- Standard errors: HC3 heteroskedasticity-robust

Italy lacks a comparable exogenous regulatory shock, making a quasi-causal ITS design infeasible. We therefore **rely on cross-sectional OLS with fixed effects** to identify robust associations.

MAIN RESULT (ITALY)

Airbnb presence is positively associated with rents

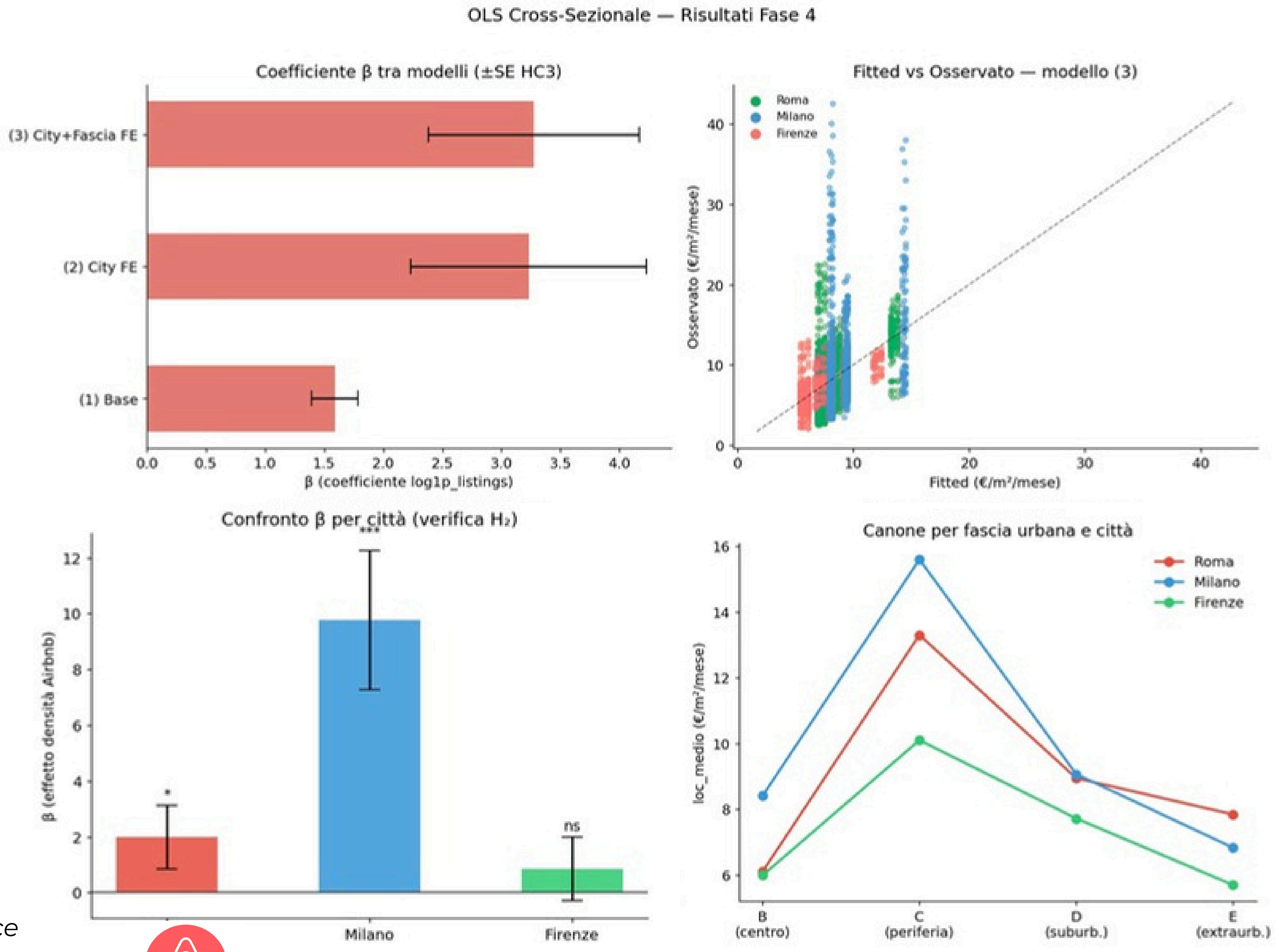
For Italian cities, we find a **strong and statistically significant positive relationship between Airbnb listings and residential rents**.

In the main specification, a doubling of listings is associated with an **increase of approximately 159 euros per month for a standard 70-square-meter apartment**.

However, this **effect is not uniform across cities**. It is strongest in Milan, weaker in Rome, and not statistically significant in Florence. This suggests that **local market conditions play an important role**.

It is important to emphasize that this result reflects **correlation rather than strong causal identification**.

H1 ✓ : Positive and significant association confirmed
H2 ✓ (partial) : Effect strongest in Milan, not significant in Florence



SPATIAL

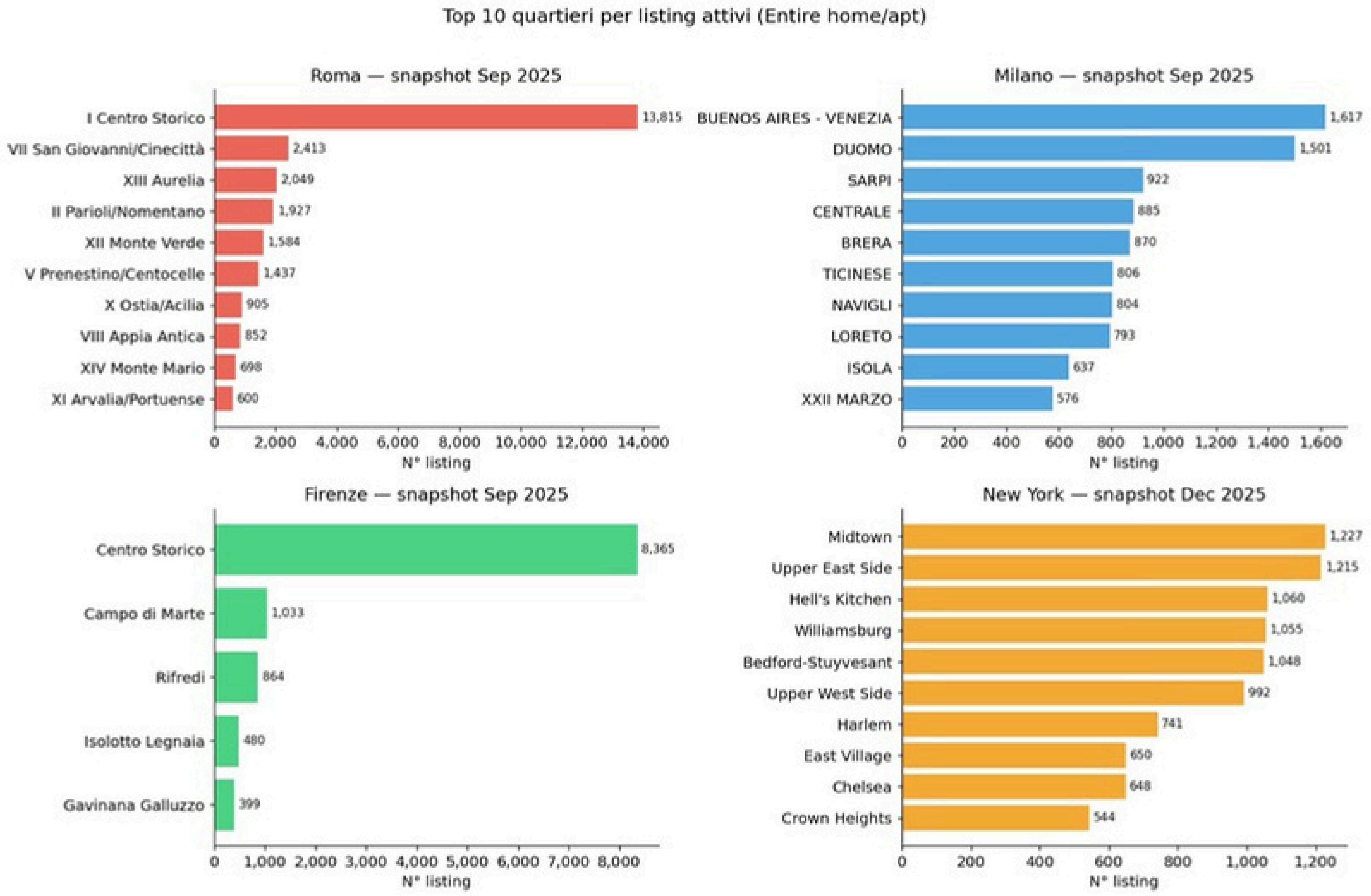
PATTERNS

Airbnb **listings are not well distributed** across cities.

Florence shows a strong concentration in the historic center, while Rome is more dispersed. Milan and New York display multiple clusters linked to economic and tourist activity.

This **spatial concentration reinforces the idea that Airbnb affects specific local housing markets rather than entire cities uniformly**.

- **Florence**: strong concentration in historic center
- **Rome**: more dispersed
- **Milan/NYC:** multiple hotspots



SYNTHESIS

Despite relying on different methodologies and data structures, the two analyses lead to a consistent conclusion.

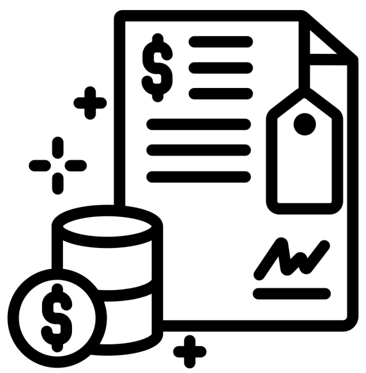
In New York, **a sudden reduction in Airbnb supply leads to an immediate decrease in rents.**

In Italy, a **higher presence of Airbnb listings is systematically associated with higher rent levels**.

Taken together, these results support the hypothesis that **short-term rental platforms reduce the effective housing supply for long-term residents, thereby increasing rental prices.**

POLICY IMPLICATIONS

The findings suggest that short-term rental platforms have measurable effects on housing markets and therefore **require careful regulation**.



Policies such as **host registration systems**, limits on short-term rentals in high-demand areas, and **targeted taxation** may help mitigate the impact on housing affordability.

At the same time, the **heterogeneity across cities indicates that regulation should be context-specific** rather than uniform.

<u>Airbnb is not only a tourism platform: it is also a housing market force.</u>

LIMITATIONS

NYC

ISSUES:

- Small sample (N = 34)
- Post-trend sensitive to specification

FUTURE FIX:

- longer post-period data, post-trend sensitive to speculation

ITALY

ISSUES:

- Endogeneity concerns
- Limited within-city variation
- Coefficient mostly identified between cities

FUTURE FIX:

- Panel data with within-city time variation

The New York analysis is based on a **relatively small sample** and involves **structural multicollinearity** inherent in the ITS design. While the **immediate effect is robust, longer-term dynamics are more uncertain**.

For Italy, the main limitation is identification. The variation in Airbnb listings is largely across cities rather than within cities over time, which limits causal interpretation.

These limitations suggest that the results should be interpreted as strong evidence of a relationship, but not as definitive proof of causality in all contexts.

THE AIRBNB EFFECT

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Thank you for your attention!